Project Proposal Guidelines

The project proposal is an essential first step in scientific work, guiding students to engage deeply with a specific topic. This initial outline encourages students to generate their own ideas while obtaining valuable early-stage feedback from the course instructor and researchers in the field. Such feedback helps refine the research question, allowing for clearer, more precise project aims. Additionally, the proposal checks students' assumptions, potentially identifying methodological missteps early on.

The proposal should be accessible to educated readers who may not specialize in the specific area of research, ensuring the clarity of main ideas and the importance of the subject matter. At the same time, it should contain enough technical detail to convince experts of the feasibility and practical planning of the proposed work. No results are expected at this stage.

Proposal structure:

1. **Introductory Paragraph**
    - **Problem Statement and Importance:** Clearly outline the problem addressed by the project, emphasizing its relevance in the broader context of engineering and computational methods. Describe why this problem is significant for engineering applications or theory and how solving it could contribute to advancements in the field.
    - **Scope of the Project:** Narrow the focus by explaining the specific scope within the broader problem. Define which aspects of the problem will be explored and why this particular focus has been chosen.
2. **Brief State of the Art (Literature Review)**
    - **Review of Relevant Literature:** Summarize key literature in the chosen area, highlighting significant studies and methodologies relevant to the proposed project. Include sources that present foundational theories or innovative approaches using methods like Finite Element Method (FEM), Finite Difference Method (FDM), Multibody System (MBS) simulations, Discrete Element Method (DEM), etc.
    - **Comparison of Methods:** Briefly discuss the methodologies chosen in the reviewed literature, focusing on their strengths and any limitations. Point out specific challenges that these methods have encountered in addressing similar problems.
    - **Identified** Gaps: If applicable, note drawbacks or disadvantages in existing approaches. Explain how these drawbacks may limit the applicability or accuracy of the models and why they necessitate further research.
3. **Open Research Question**
    - **Gap in Knowledge:** Describe the specific research gap that the project will aim to address. This gap could involve limitations in current simulation methods, areas where theoretical work has yet to align with empirical data, or complex phenomena that remain unexplained.

- **Unsolved Challenges**: Outline existing challenges or unanswered questions within the field. Discuss potential reasons why these problems remain unresolved, whether due to technological limitations, lack of refined methodologies, or gaps in interdisciplinary research.

4. **Methodology**
   - **Objectives and Techniques:** Define the project's primary objectives and the research techniques that will be employed to achieve them. This may involve designing simulation models, experimenting with different numerical methods, or developing new algorithms.
   - **Simulation Tools:** Specify potential simulation tools and software (e.g., Abaqus for FEM, MATLAB for FDM, Adams for MBS, LIGGGHTS for DEM) that may be used. Provide a brief justification for each chosen tool, linking it to the unique requirements of the project.
   - **Anticipated Limitations:** Acknowledge possible limitations and challenges, including computational constraints, data availability, or potential inaccuracies in simulation models. Outline any preliminary plans for managing or mitigating these issues.
   - **(Resource Requirements:** If applicable, detail any special resources that may be required, such as specific materials, equipment, or advanced software packages. Consider resources that will support the accurate or efficient execution of simulations.**)**
5. **Outcome**
   - **Planned Results:** Describe the expected technical outcomes of the project, such as improved simulation accuracy, validation of a novel approach, or refined model parameters. Explain how these results align with the initial objectives.
   - **Contribution to Knowledge**: Explain how the anticipated results will enhance understanding in the field, either by confirming theoretical models, providing more precise engineering tools, or opening new avenues for practical application.
6. **Conclusion**
   - **Summary and Next Steps:** Summarize the project's aims, intended contributions, and methodological approach. Indicate the next steps in the research process, which might include developing a detailed project timeline, securing necessary resources, or preparing a more comprehensive literature review.